



Improving Wireless Charging Efficiency Using Photo-Sensing Multi- Transmitter Coil Arrays



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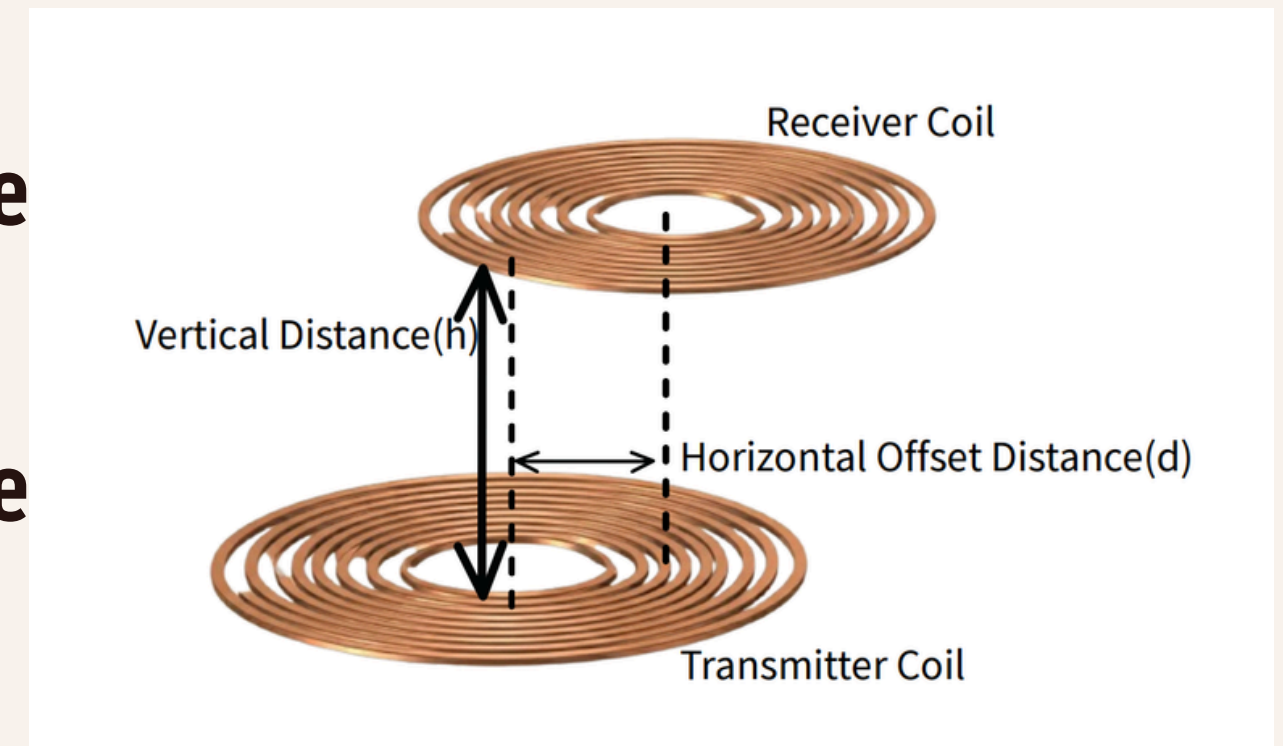
Research Motivation

The inspiration for this research stems from observing the gap in "Human-Computer Interaction" in daily life. In modern usage, users often move their bodies unconsciously while using mobile devices (e.g., tilting the device during gaming or operation). However, existing single-coil wireless charging technologies require high alignment precision. This leads to frequent power interruptions when "charging while using," which not only disrupts the user experience but also accelerates battery degradation due to repeated charging and discharging cycles.

Experimental Methods

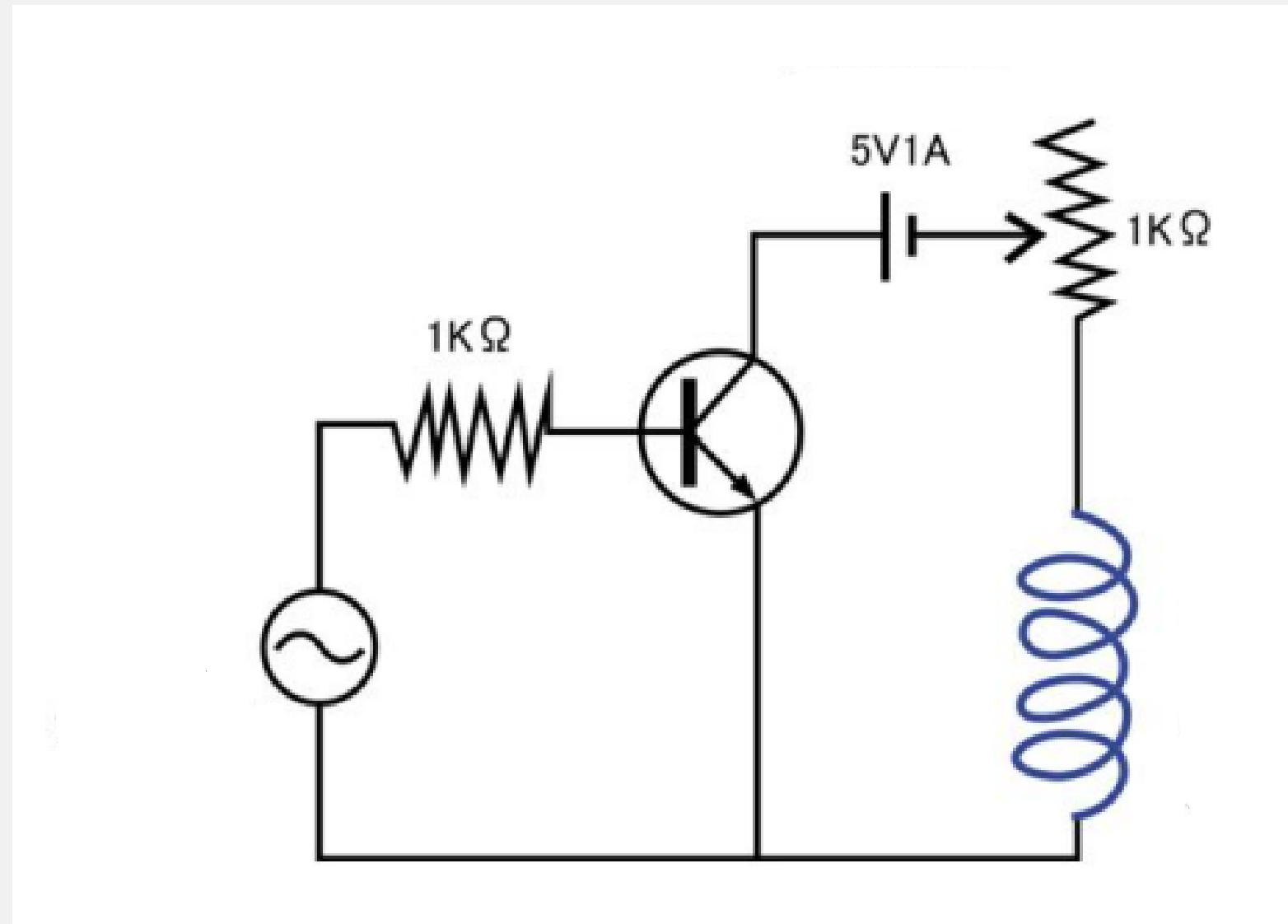
Our experiment is divided into five parts

- The effect of the vertical distance between the receiving coil and the transmitting coil on the received voltage
- The effect of the horizontal offset distance between the receiving and transmitting coils on the received voltage
- The effect of the horizontal offset distance between the two transmitting coils on the receiving voltage
- The effect of the transmitting signal frequency on the receiving voltage
- Integrate previously built experimental modules.

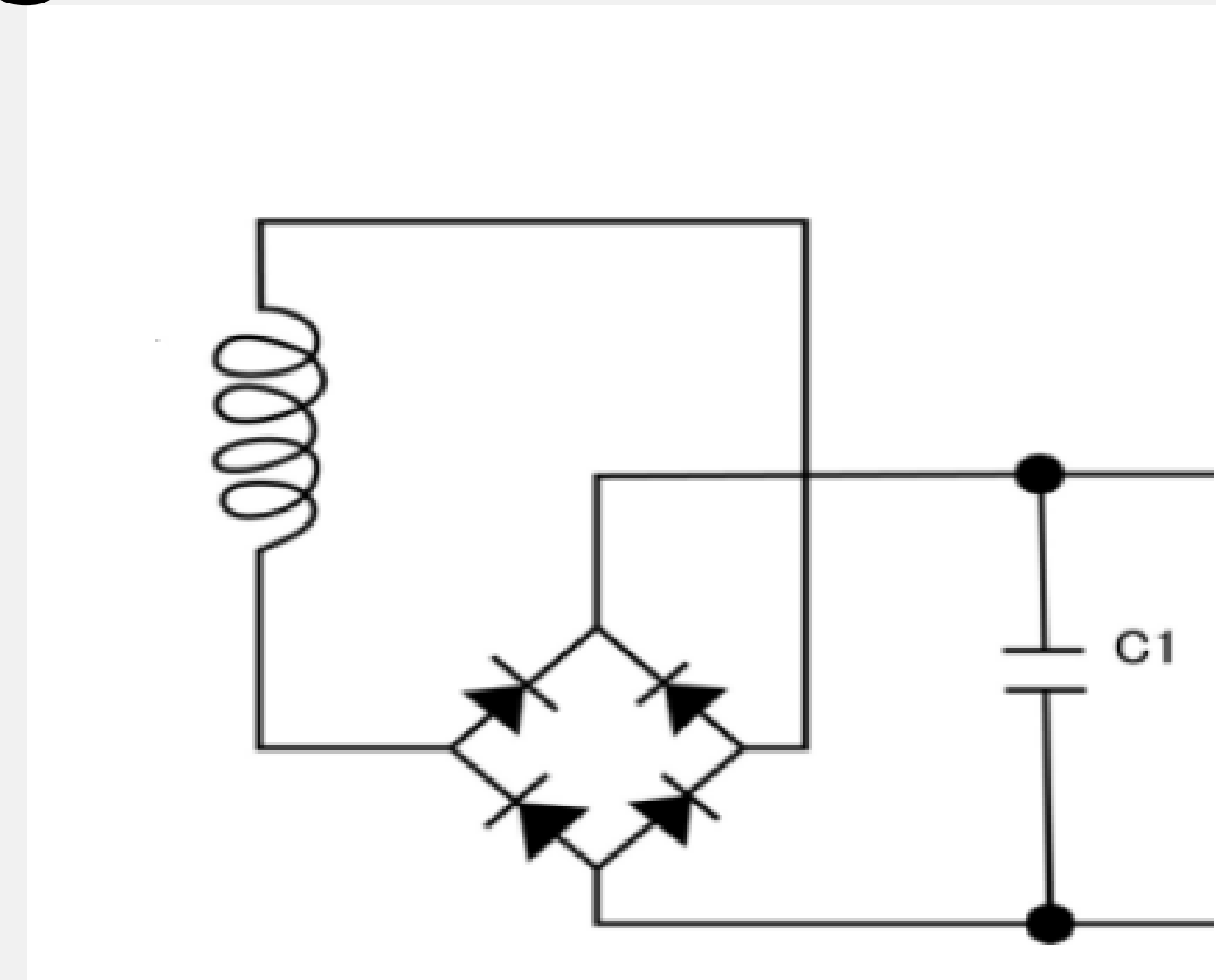


Experimental Methods

-Device circuit diagram



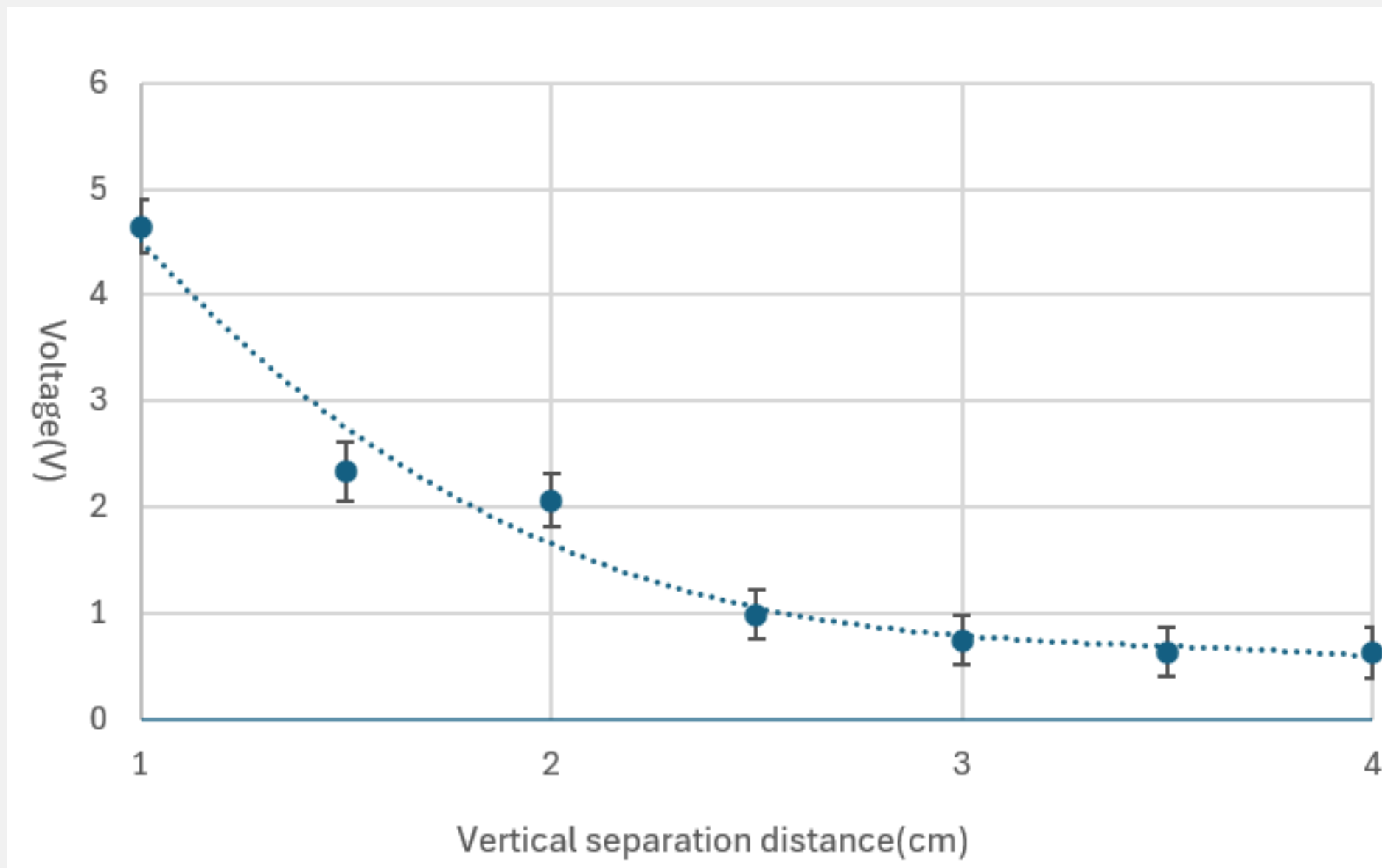
Transmitter



Receiver

Research Results

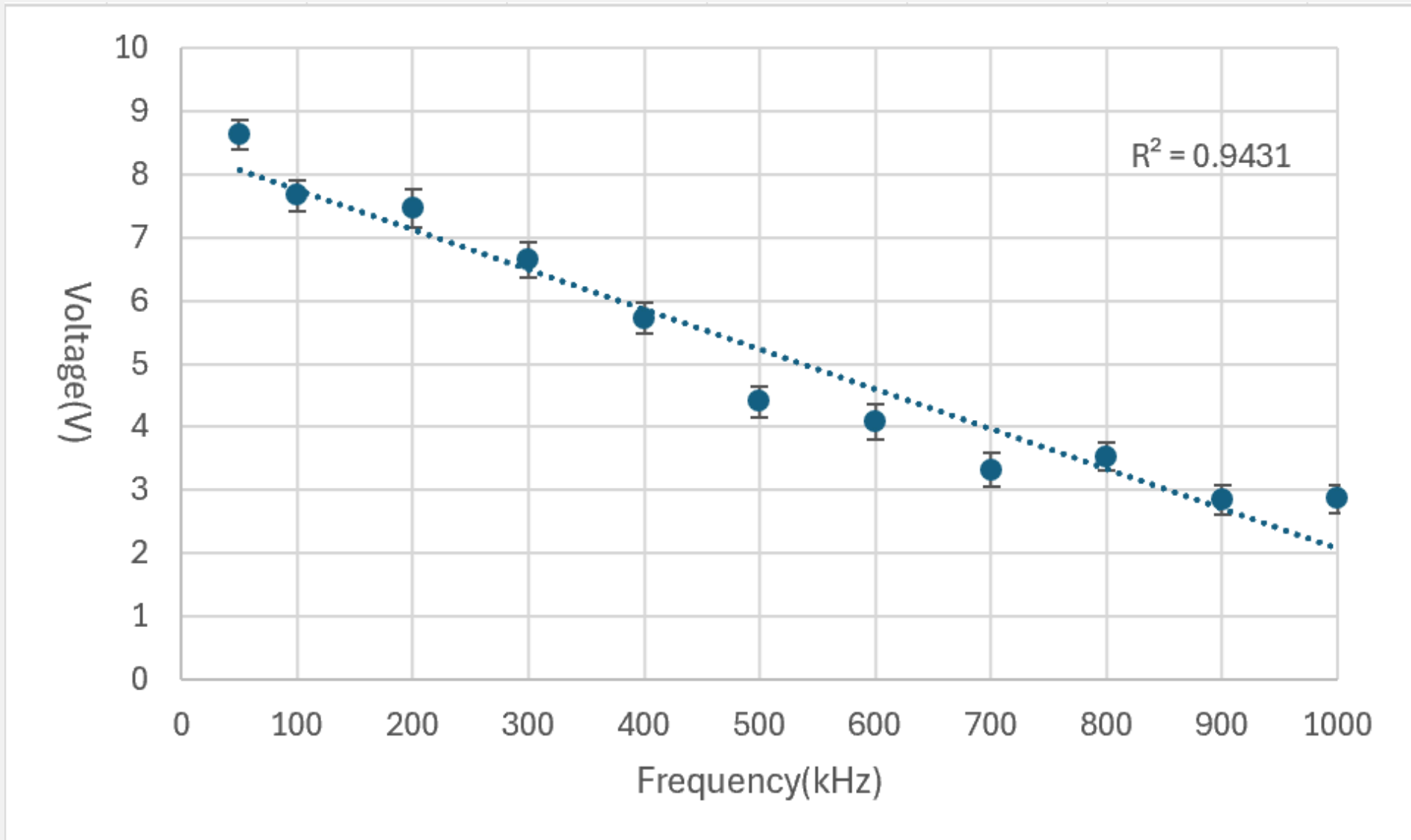
**vertical distance between
the receiving coil and the
transmitting coil**



**As shown in the figure beside,
the receiver voltage exhibits
exponential decay as the
vertical distance increases.**

Research Results

the transmitting signal frequency

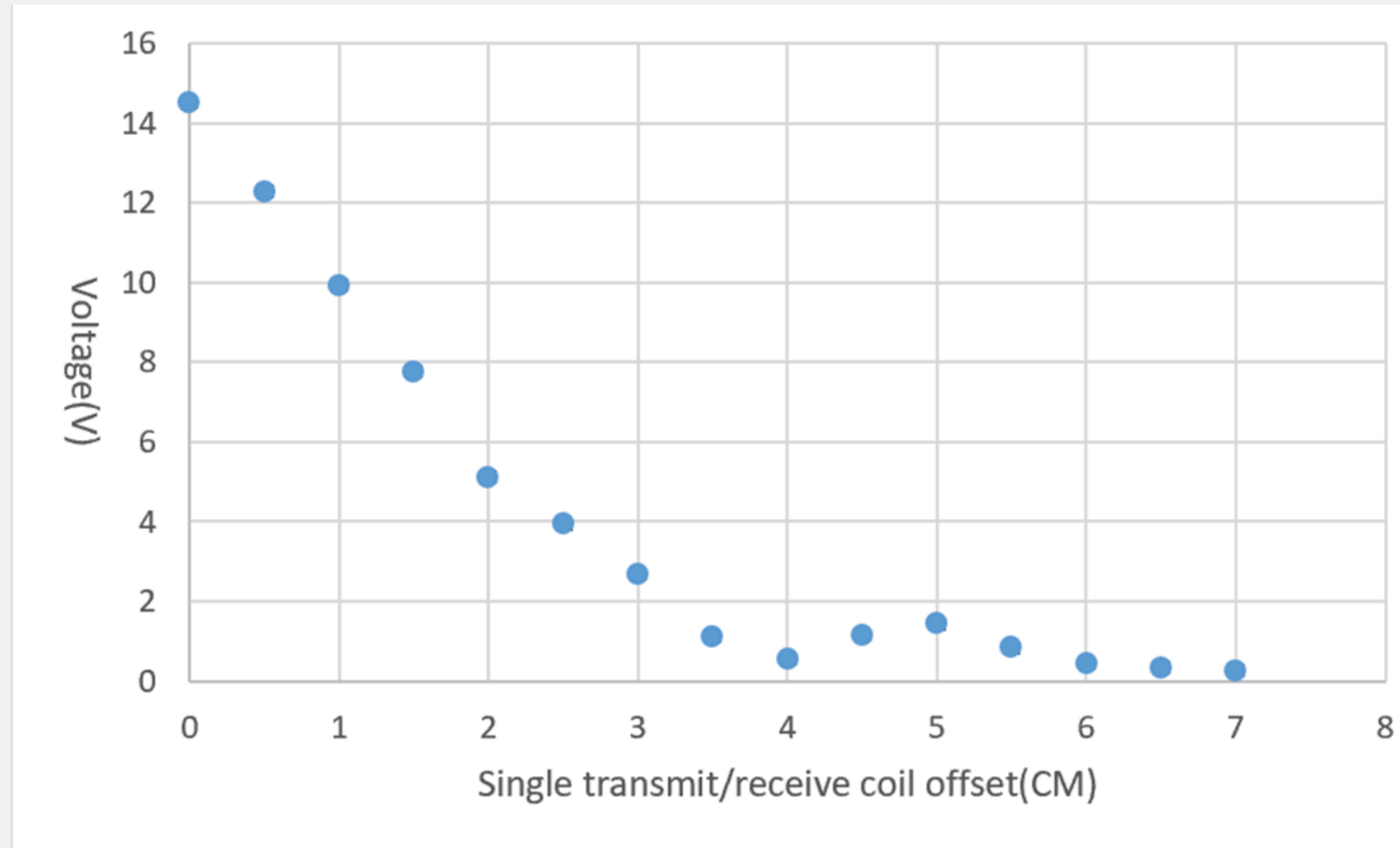


Experimental data shows that the induced voltage decreases steadily as the operating frequency increases, with a linear fit correlation coefficient of 0.9431. .

Research Results

The horizontal offset distance between the receiving and transmitting coils

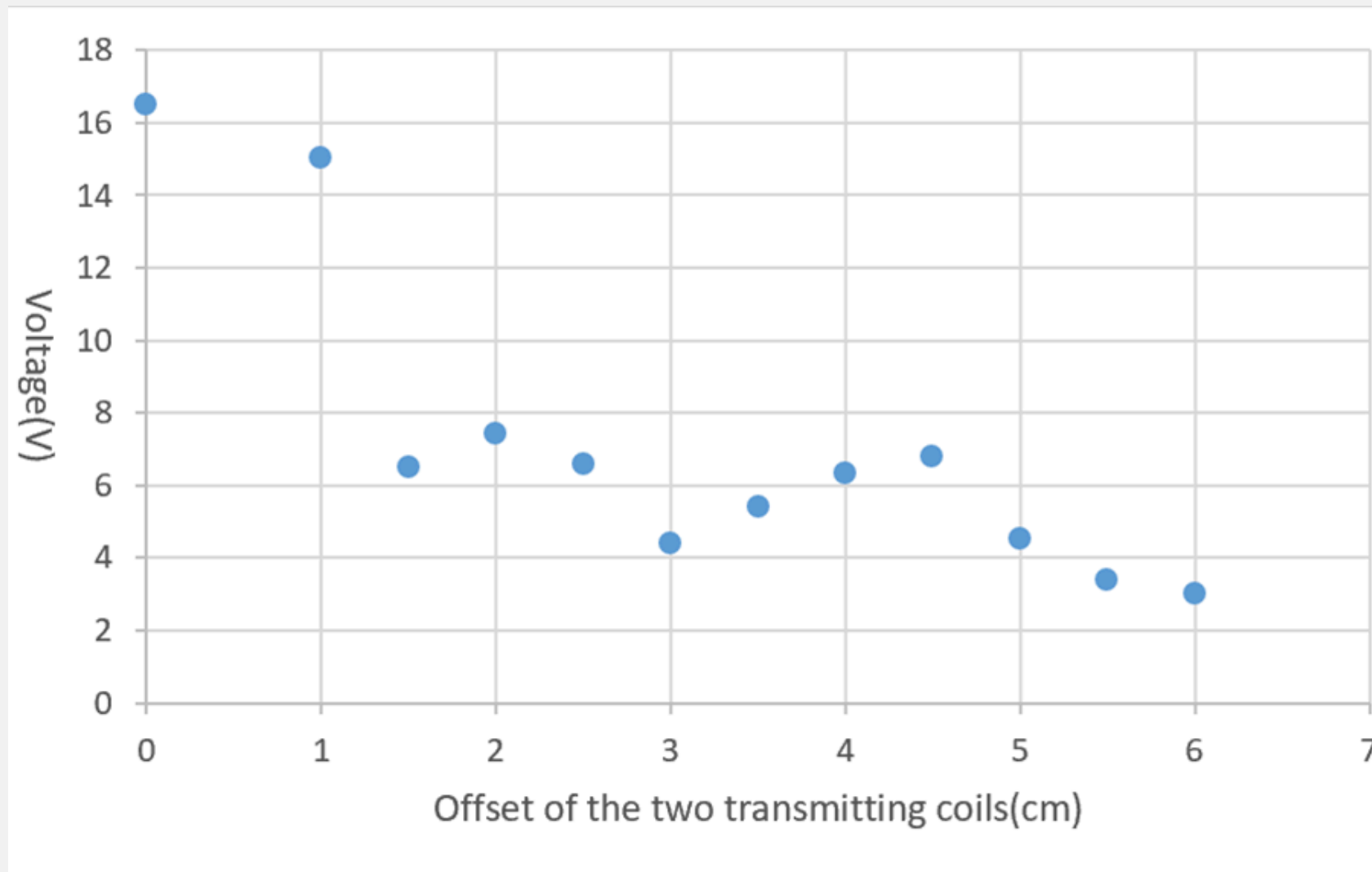
As the horizontal displacement between the transmitting and receiving coils increases, the voltage decays rapidly. Although there is a slight increase at 5cm displacement, the receiving voltage remains low.



Research Results

**the horizontal offset distance
between the two transmitting coils**

**As the horizontal displacement
between two transmitting coils
increases, the induced voltage
shows an overall downward trend.
However, local rebounds occur
near 2cm and 4.5cm, and between
1.5cm and 4.5cm, the receiver
voltage is between 4~7V. After
5cm, it continues to decrease.**



Experimental Methods

Integrate previously built experimental modules.

1. Determining Optimal Transmission Parameters

Frequency Selection: The final frequency was set at 150kHz.

Effective magnetic coupling transmission was confirmed to be maintained within a 1cm vertical distance.

2. Addressing Displacement-Induced Power Loss

A physical overlap design of 0.5–1cm between the coils was adopted.

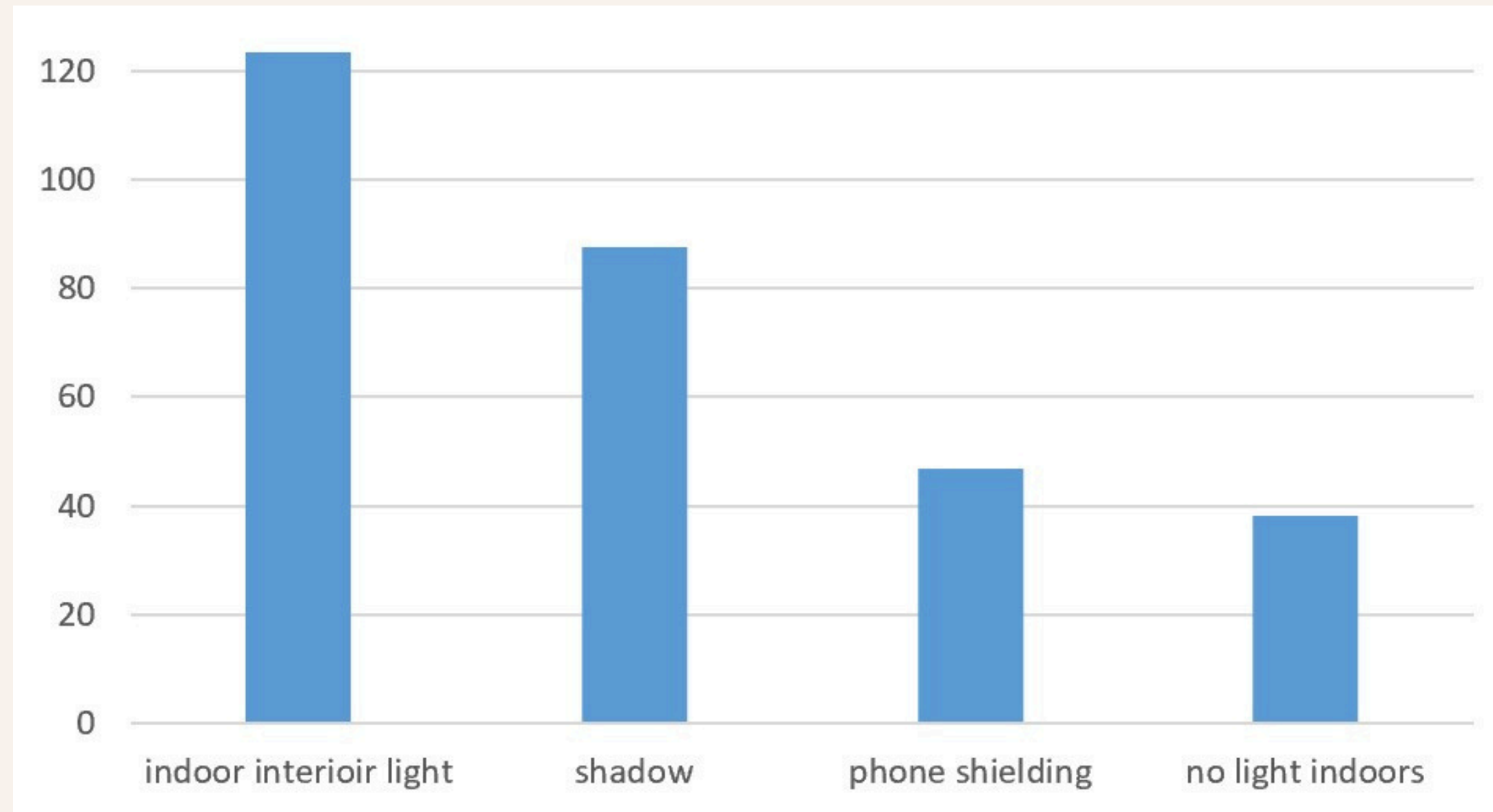
3. High-Reliability Energy-Saving Control

Optimized Optical Trigger: After testing, the coil activation threshold was locked at 50 (ambient background value approximately 120).

Comparative Advantage: Compared to values of 90 (too sensitive) or 40 (too insensitive), a value of 50 accurately activates the corresponding coil.

Experimental Methods

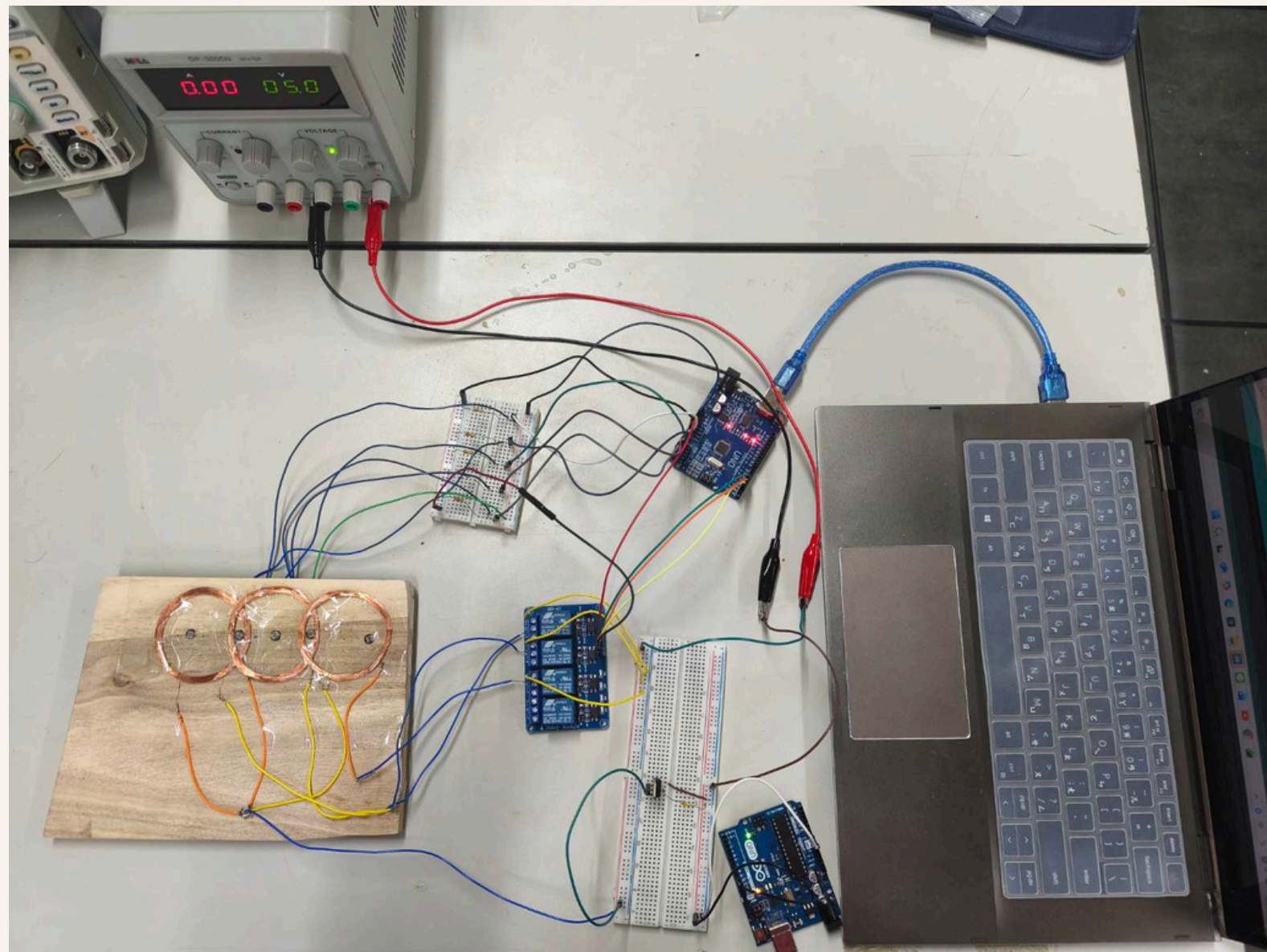
Integrate previously built experimental modules.



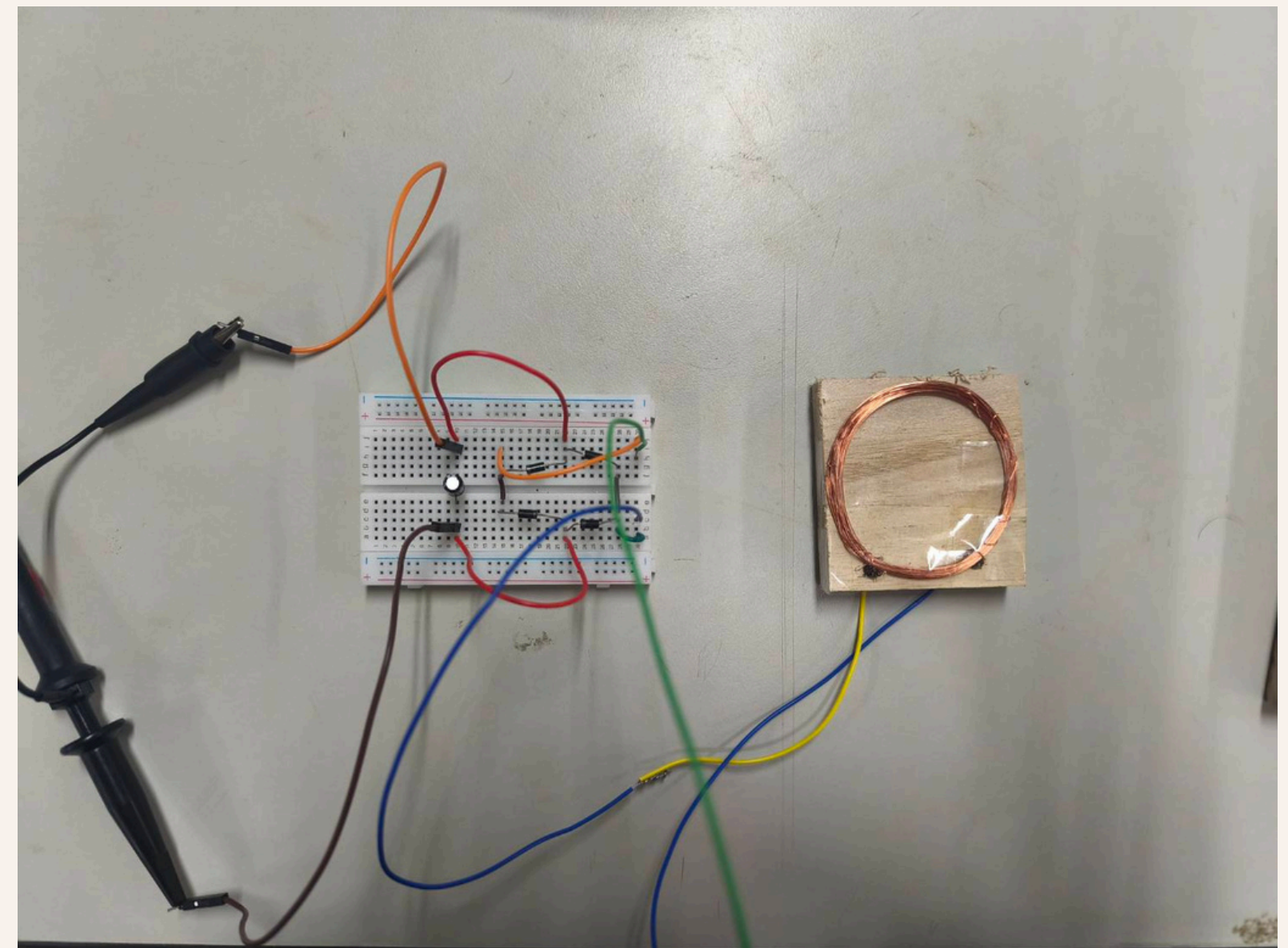
Values displayed by the photoresistor under different lighting conditions

Experimental Methods

Integrate previously built experimental modules.

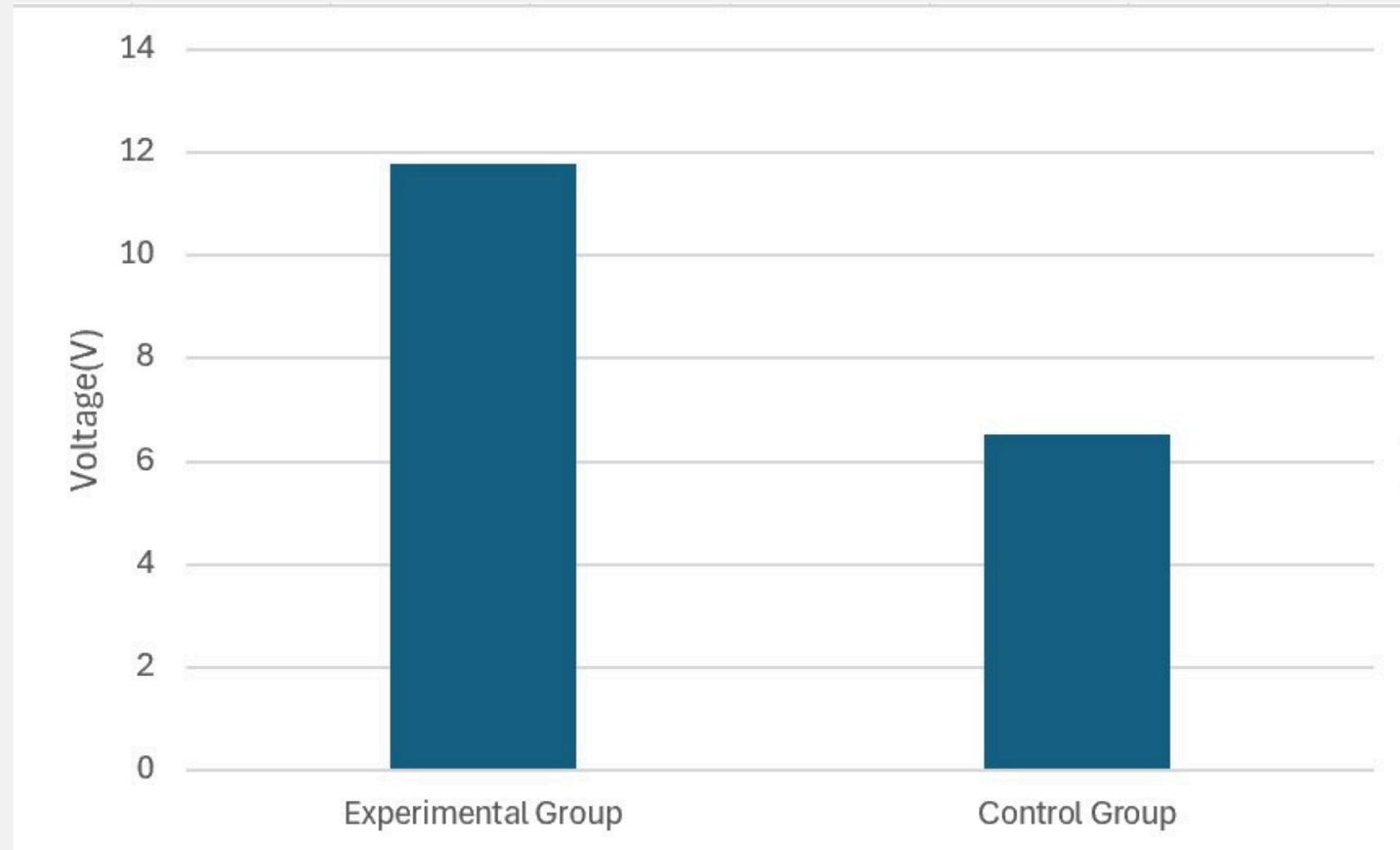


Transmitter

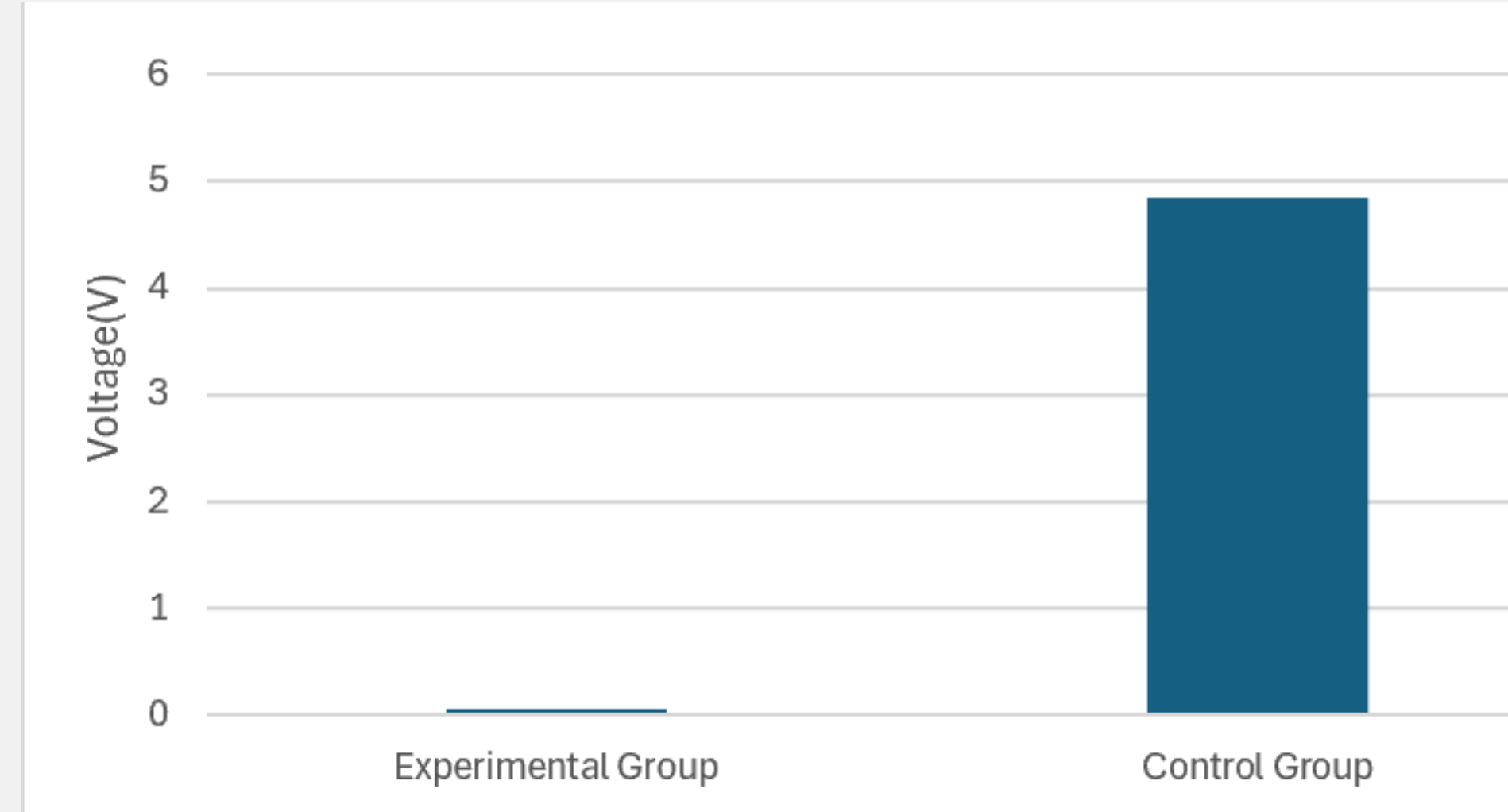


Receiver

Research Results



Comparison chart of receiver voltage between the experimental group and the control group



Comparison of no-load energy consumption between the experimental group and the control group



Thank you

